

TEAM CONTRIBUTION DOCUMENT

MarketMind AI

AI-Powered Business Intelligence and Decision Support Platform

Project Guide

Shravya

Team Members

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2. Neelam Rishika Damini
3. Pallavi D R
4. Kavya Namala
5. Pramodh Pillitla
6. Satish Mukkidi

Infosys Springboard

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Chapter 1

Project Information

- **Project Title:** MarketMind AI
- **Project Description:** AI-Powered Business Intelligence and Decision Support Platform.
- **Project Guide:** Springboard Mentor
- **Team Members:**
 1. Susanna Dontha
 2. Neelam Rishika Damini
 3. Pallavi D R
 4. Kavya Namala
 5. Pramodh Pillitla
 6. Satish Mukkidi

Chapter 2

Introduction

This document presents the individual contributions of the team members who worked on the MarketMind AI project across the different project milestones.

MarketMind AI is an AI-powered business intelligence and decision support platform integrating frontend and backend technologies, database services, data processing, machine learning models, business analytics, and visualisation features.

The document records the primary responsibilities and contributions of each team member across the milestones, along with the major challenges encountered and the approaches used to address them.

The contributions are organised according to the project milestones to provide a clear record of the work completed by each team member.

Team Members and Contribution Areas

The MarketMind AI project was developed collaboratively through frontend development, backend services, AI/ML implementation, system integration, testing, documentation, and deployment activities.

The following table summarises the primary contribution areas of each team member.

No.	Team Member	Primary Contribution Areas
1	Neelam Rishika Damini	Frontend development and UI/design, including role-based dashboard design and implementation with navigation and user permissions, guided workflows, chatbot-related functionality, authentication, error handling and validation, schema-related work.
2	Pallavi D R	Frontend and backend development, implementation of role-based users, application integration, implementation of churn prediction, visualization components, and user interface improvements and optimization .
3	Susanna Dontha	AI/ML development with work on customer segmentation, sales forecasting, and product recommendation functionality. Contributed to frontend-backend and API integration, testing, error handling, system design improvements, final project documentation.
4	Kavya Namala	AI/ML and data-driven module development, including revenue prediction, business analytics, data processing, and analytical workflows.
5	Pramodh Pillitla	Backend development, API integration, backend services, database and system integration, deployment, Git/GitHub issue resolution, and overall application functionality.
6	Satish Mukkidi	Backend development, integration support, system functionality, and Git-related collaboration.

Chapter 3

Individual Contributions

3.1 Susanna Dontha

Susanna Dontha contributed to the development of MarketMind AI across multiple stages of the project. Her work focused on dataset preparation and preprocessing, AI/ML development, customer segmentation, sales forecasting, product recommendation, frontend and backend integration, API integration, testing, error handling, Git and GitHub collaboration, and project documentation.

3.1.1 Milestone 1 Contributions

During the initial phase of the project, Susanna contributed to establishing the foundation required for the AI/ML and application components.

- Worked with project datasets related to sales, customers, products, inventory, and business transactions.
- Contributed to data cleaning and preprocessing activities, including handling missing values, duplicate records, inconsistent data formats, and preparing relevant data for analysis.
- Worked on understanding dataset structures, attributes, and relationships to support future analytics and AI/ML modules.
- Assisted in identifying relevant features and attributes required for customer analysis, sales forecasting, customer segmentation, and recommendation-related workflows.

- Contributed to exploratory analysis of the available business data to identify patterns and relationships that could support the development of AI/ML features.
- Assisted in preparing and validating processed datasets for use in subsequent analytics, machine learning, and recommendation workflows.

3.1.2 Milestone 2 Contributions

During Milestone 2, Susanna contributed to the development and integration of AI/ML functionality and business intelligence workflows within the MarketMind AI application.

- Worked on AI/ML-related components, including customer segmentation, sales forecasting, and business intelligence workflows.
- Contributed to customer segmentation by working with customer-related data and identifying meaningful characteristics that could be used for grouping customers.
- Worked on analysing customer behaviour and transaction-related data to support customer segmentation and business insights.
- Contributed to sales forecasting workflows by preparing sales data and supporting the prediction of future sales trends.
- Worked on data preprocessing and feature preparation required for machine learning and predictive analytics workflows.
- Improved the accuracy and reliability of AI/ML outputs through data validation, feature refinement, model evaluation, and iterative testing of the implemented workflows.
- Optimised AI/ML workflows by refining data processing and model execution steps to improve efficiency and reduce unnecessary processing.
- Assisted in integrating AI/ML functionality with the backend so that processed data and model outputs could be accessed through application services and APIs.
- Worked on frontend–backend integration to ensure that AI/ML outputs and business insights could be displayed within the user interface.
- Contributed to API integration between frontend and backend components.
- Assisted with implementing and improving frontend features related to analytics, business insights, and AI-powered functionality.
- Participated in testing integrated workflows to verify that frontend requests, backend APIs, and AI/ML processing were functioning together correctly.

3.1.3 Milestone 3 Contributions

During Milestone 3, Susanna focused extensively on the development and integration of the Product Recommendation System and related advanced application features.

- Worked on the development and integration of the Product Recommendation System within the MarketMind AI platform.
- Contributed to analysing customer, product, and transaction-related information required for generating meaningful product recommendations.
- Worked on preparing and processing data used by the recommendation workflow.
- Assisted with the development and refinement of recommendation logic to support personalized and relevant product suggestions.
- Worked on the frontend implementation and integration of recommendation-related features.
- Contributed to displaying recommendation outputs and related information within the application's user interface.
- Worked on connecting frontend recommendation features with backend APIs and recommendation-processing workflows.
- Assisted with backend integration required to process requests and return recommendation results to the frontend.
- Participated in evaluating AI/ML-related workflows using appropriate evaluation approaches and performance metrics where applicable.
- Worked with evaluation concepts and metrics such as Precision, Recall, F1 Score, and other relevant validation measures for assessing machine learning and prediction-related workflows.
- Assisted with validating model behaviour and analysing outputs across different data scenarios.
- Tested recommendation functionality and verified that outputs were correctly generated and displayed through the application.
- Contributed to resolving Git and GitHub-related issues encountered during collaborative development, including branch synchronization and integration-related issues.

3.1.4 Milestone 4 Contributions

During the final milestone, Susanna contributed to testing, error handling, Git-related tasks, and the preparation of the final project deliverables.

- Worked on testing different modules and workflows of the MarketMind AI application.
- Participated in functional testing of implemented features to verify that the frontend, backend, APIs, datasets, and AI/ML components were functioning correctly.
- Tested AI/ML-related workflows, including customer segmentation, sales forecasting, and product recommendation functionality.
- Worked on validating application behaviour using different data scenarios and inputs.
- Assisted in identifying system issues, integration problems, and errors encountered during testing.
- Worked on error handling and contributed to identifying and resolving issues related to application workflows and frontend–backend communication.
- Assisted with resolving integration and communication issues, including CORS-related issues affecting frontend–backend connectivity.
- Assisted with resolving Git and GitHub-related issues and reviewing repository changes during the final stages of development.
- Took primary responsibility for preparing and organizing the Final Project Documentation according to the structure and instructions provided by the project mentor.
- Documented major project components, including the technology stack, system architecture, AI/ML implementation, database and APIs, user interface, GitHub and version control, testing, results, deployment, challenges, future scope, conclusion, and references.
- Worked on documenting AI/ML modules, including datasets, preprocessing workflows, model-related implementation, evaluation approaches, and AI/ML workflows.
- Collected and organized screenshots, diagrams, system outputs, and other visual materials required for the final project documentation.

- Reviewed and reorganized the documentation to improve clarity, consistency, and alignment with the required documentation structure.
- Prepared the separate Team Contribution Documentation requested by the mentor.
- Coordinated with team members to collect information regarding their individual contributions, responsibilities, challenges encountered, and solutions implemented.
- Organized and compiled the individual team member contributions into a structured contribution document.

3.1.5 Challenges and Resolutions

Git and Branch Synchronization Issues

During collaborative development, issues related to branch synchronization, integrating changes, and managing updates across different project components were encountered. These issues were addressed by reviewing repository changes, synchronizing branches, resolving integration problems, and coordinating changes before incorporating them into the shared project repository.

Frontend–Backend Communication and CORS Issues

Integration-related issues affecting communication between the frontend and backend were encountered during development and testing. These issues were investigated by checking API communication, application configuration, and data flow between components. CORS-related configuration and integration issues were addressed to improve frontend–backend connectivity.

AI/ML Workflow Validation

Validating AI/ML workflows across different datasets and input scenarios required additional testing and analysis. The workflows were tested using different scenarios, model outputs were reviewed, and integration between datasets, backend services, APIs, and frontend components was validated.

Final Documentation Coordination

Preparing the final documentation required collecting technical information, screenshots, diagrams, outputs, and contribution details from different project components and team members. This was addressed by organizing the documentation according to the mentor’s required structure and coordinating with team members to collect the necessary information.

3.2 Neelam Rishika Damini

Neelam Rishika Damini contributed extensively to the full-stack development, integration, testing, and enhancement of the MarketMind AI platform. Her work included frontend and backend development, authentication and role-based access control, dashboard and CRUD functionality, AI/ML page integration, advanced platform features, security improvements, bug fixing, testing, and documentation.

3.2.1 Milestone 1 Contributions

During the initial development phase, Rishika contributed to establishing the core application architecture and implementing major frontend and backend features.

- Contributed to the initial React frontend setup using Vite and the configuration of the frontend project structure.
- Participated in setting up the FastAPI backend, SQLAlchemy ORM, and PostgreSQL database integration.
- Worked on organizing the project structure, including frontend components and backend modules such as routers, models, schemas, and application services.
- Developed and integrated authentication-related features, including login and registration interfaces.
- Worked on connecting frontend authentication functionality with backend authentication APIs.
- Implemented role selection during user registration for different application roles, including Business Owner, Store Manager, Sales Executive, and Administrator.
- Contributed to role-based access control and protected route functionality to restrict access to pages based on user roles.
- Worked on authentication and session-related improvements, including persistent token handling.
- Contributed to the development and enhancement of major application interfaces, including dashboards and business management pages.
- Worked on features related to sales tracking, inventory monitoring, customer management, invoices, categories, suppliers, and team management.

- Contributed to customer-related analytics and AI-powered pages, including customer segmentation and churn-related functionality.
- Assisted in integrating AI/ML pages and workflows related to sales forecasting, customer segmentation, and churn prediction.
- Contributed to the development and enhancement of the landing page, including responsive design, theme-related features, and interactive user interface improvements.
- Assisted with resolving issues related to authentication, protected routes, database connectivity, empty data handling, and AI page crashes.

3.2.2 Milestone 2 Contributions

During Milestone 2, Rishika focused on advanced application functionality, user experience improvements, AI integration, and collaboration across different project modules.

- Developed activity and audit-related functionality to support the tracking of user actions within the application.
- Contributed to implementing global search functionality to improve navigation and accessibility across different application pages.
- Added loading-related improvements, including skeleton loading interfaces and improved feedback during data retrieval.
- Worked on user feedback features such as toast notifications.
- Contributed to the enhancement of the AI chatbot by improving responses and adding interactive features.
- Worked on chatbot functionality involving natural language responses, text-to-speech support, and a business-focused knowledge base.
- Implemented and enhanced guided tour functionality to help users understand different features of the application.
- Added tour persistence and replay functionality through user preferences and application settings.
- Worked on improving routing and navigation for newly added and updated application pages.

- Contributed to dashboard improvements, including loading states and handling situations where new users did not initially have sufficient dashboard data.
- Assisted in improving role-based access control across multiple application routes.
- Participated in integrating AI/ML modules developed by different team members into the main application.
- Assisted with resolving merge conflicts and integration issues that occurred when multiple team members worked on overlapping project files.
- Worked on validating and improving anomaly detection, churn prediction, and other integrated AI/ML workflows.

3.2.3 Milestone 3 Contributions

During Milestone 3, Rishika contributed to advanced platform features, real-time functionality, reporting, security, accessibility, and user experience improvements.

- Contributed to the implementation of multilingual application support and language-switching functionality.
- Worked on Progressive Web Application (PWA) functionality, including application metadata, offline support, and installation-related features.
- Contributed to real-time alert functionality using WebSocket-based communication.
- Worked on alert broadcasting and user-facing live notification functionality.
- Developed and enhanced reporting-related features, including multiple report templates and export functionality.
- Contributed to PDF and spreadsheet export workflows for business reports and analytics.
- Worked on scheduled reporting functionality, including report configuration and frequency selection.
- Contributed to dashboard customization and dashboard layout functionality.
- Worked on user security features, including two-factor authentication and improvements to security-related settings.
- Contributed to user onboarding functionality and guided setup workflows.

- Worked on audit and compliance-related features for tracking application activities.
- Implemented and enhanced user experience features, including command palette functionality, error boundaries, QR code generation, and responsive interface improvements.
- Assisted in identifying and resolving issues related to scheduled reports, installation functionality, dashboard loading, and multilingual application behaviour.

3.2.4 Milestone 4 Contributions

During the final milestone, Rishika focused on bug fixing, application testing, security hardening, responsiveness, error handling, and production-readiness improvements.

- Worked on improving sales data handling and pagination to support large datasets more efficiently.
- Contributed to backend security improvements, including environment configuration for sensitive values and configurable application settings.
- Worked on improving CORS configuration and frontend–backend communication.
- Assisted in resolving backend routing and activity logging issues.
- Contributed to improving application error handling by replacing silent error handling patterns with meaningful logging.
- Worked on database connectivity and connection reliability improvements.
- Assisted in resolving frontend crashes caused by changes in backend data formats.
- Worked on improving loading states, empty states, and frontend error handling across multiple application pages.
- Contributed to fixing CORS-related browser communication issues, including support for different local development origins.
- Worked on improving mobile responsiveness, including mobile sidebar navigation and responsive layout behaviour.
- Contributed to user interface improvements, including animated KPI displays, loading placeholders, and visual refinements.

- Participated in comprehensive application testing across different user roles, APIs, CRUD operations, role-based access restrictions, database isolation, and AI/ML endpoints.
- Assisted in verifying that integrated AI/ML modules and application features were functioning correctly under different user roles and scenarios.
- Contributed to updating project documentation and configuration files required for application setup.

3.2.5 Challenges and Resolutions

Authentication and Role-Based Access Issues

Authentication and authorization issues were encountered during the integration of frontend and backend components. These included token persistence and restricted page access. The issues were addressed by improving token handling, protected route implementation, and role-based access checks.

AI/ML Integration and Empty Data Handling

Some AI-related pages experienced issues when data was unavailable or when integrated model outputs were incomplete. These issues were addressed by adding appropriate validation, fallback behaviour, empty-state handling, and improved integration between the AI/ML modules and application components.

Merge Conflicts and Multi-Developer Integration

Multiple team members worked on interconnected files and modules, resulting in merge conflicts during integration. These issues were resolved through manual review of conflicting changes, careful integration of the required functionality, and coordination between team members.

Frontend and Backend Data Compatibility

Changes to backend response formats caused compatibility issues with some frontend pages. The affected components were updated to correctly handle the revised data structures and loading behaviour.

CORS and Application Connectivity

Browser requests were affected by CORS configuration issues during local development and testing. The application configuration was updated to allow the required frontend development origins and improve communication between the frontend and backend.

Application Stability and Error Handling

Several issues related to missing routes, loading states, frontend crashes, and silent error handling were identified during testing. These were addressed by improving logging, adding error boundaries, correcting application routes, and implementing better fallback and loading behaviour.

3.3 Namala Kavya

Namala Kavya contributed to MarketMind AI in the areas of dataset preparation, inventory preprocessing, revenue prediction, business-rule alerts, testing, troubleshooting, and Git-based project integration.

3.3.1 Milestone 1 Contributions

During Milestone 1, Kavya focused on preparing and preprocessing business datasets, particularly inventory-related data, to establish a reliable and structured foundation for analytics and future application development.

- Worked with inventory and other business-related datasets used in MarketMind AI and reviewed their structure before preprocessing.
- Examined dataset columns, data types, and overall data quality to understand the available information and identify potential issues.
- Developed inventory preprocessing workflows using Python and Pandas.
- Performed data quality checks for missing values, duplicate records, incorrect data types, inconsistent formats, and incomplete information.
- Conducted descriptive and exploratory analysis to better understand inventory-related variables and their overall characteristics.
- Reviewed unusual values and potential outliers to identify records requiring further inspection before being used in analytics workflows.
- Processed and standardised date-related fields into suitable formats for further analysis and application-level use.
- Cleaned and organised relevant inventory data to improve consistency and usability across subsequent project modules.

- Prepared a cleaned inventory dataset for use in analytics, predictive workflows, and future application development.
- Organised preprocessing scripts, documentation, dependency files, and cleaned dataset outputs to maintain a structured development workflow.
- Performed validation checks after preprocessing to ensure that the cleaned inventory data maintained the required structure and contained consistent and usable values.
- Identified and handled data inconsistencies that could affect inventory analysis, reporting, and downstream analytics workflows.
- Prepared relevant inventory attributes and derived fields required for subsequent business analytics and predictive modelling activities.
- Documented the preprocessing steps and data preparation process to make the workflow reproducible and easier to integrate with subsequent project modules.

3.3.2 Milestone 2 Contributions

During Milestone 2, Kavya primarily focused on developing the Revenue Prediction functionality, analysing business-related data, and preparing the predictive workflow for application-level use.

Revenue Prediction

- Developed a machine learning workflow for predicting business revenue using sales and business-related data.
- Used relevant features such as sales quantity, pricing, product category, regional information, demand, promotional data, and seasonal factors.
- Performed data preprocessing and prepared numerical and categorical variables using appropriate encoding techniques.
- Identified relevant features and prepared analytical data for model development, training, and evaluation.
- Developed and trained a Random Forest-based regression model for revenue prediction.
- Evaluated the model using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

- Prepared the trained model and preprocessing components for application-level predictions.
- Contributed to integrating the revenue prediction workflow with the backend and making the functionality accessible through the application interface.

Business Analytics Support

- Performed exploratory analysis of business-related data to support the development of the revenue prediction workflow.
- Analysed sales trends and relevant business attributes, including product information, pricing, regional data, and transaction-related variables.
- Examined the relationship between key business factors and revenue outcomes to identify relevant features for predictive analysis.
- Reviewed data distributions and basic statistical characteristics to better understand the behaviour of revenue-related variables.
- Reviewed prediction outputs to assess whether generated revenue estimates were reasonable across different business scenarios.

3.3.3 Milestone 3 Contributions

During Milestone 3, Kavya focused on developing business-rule monitoring and alert functionality, integrating the implemented features into the application, and managing project contributions through Git-based version control.

Business-Rule Alerts

- Developed a business-rule alert module separate from the machine-learning-based anomaly detection functionality.
- Implemented predefined rules for identifying unusually large sales transactions and significant stock depletion.
- Used business thresholds to identify conditions requiring attention.
- Configured alert severity levels based on the seriousness of identified business conditions.

- Integrated the alert functionality with the application's existing notification workflow.
- Tested the alert system using normal, high-risk, and critical scenarios to verify alert generation and severity classification.

Git and Project Integration

- Used the `Namala-kavya` branch for individual development and contributed to the shared `pre-dev` branch for collaborative integration.
- Pushed preprocessing, revenue prediction, and business-alert-related work to the project repository.
- Worked with the shared repository structure and reviewed differences between individual branches and integrated project versions.
- Followed Git-based version control practices during collaborative development and feature integration.
- Reviewed branch versions and integration status when functionality appeared missing or inconsistent between development branches.

3.3.4 Milestone 4 Contributions

During Milestone 4, Kavya focused on testing and validating the implemented workflows, troubleshooting application and environment issues, and verifying the integrated functionality of the project.

- Tested inventory preprocessing outputs to verify the consistency and usability of prepared datasets.
- Tested the revenue prediction functionality and verified the trained model and preprocessing components during application-level prediction.
- Reviewed prediction outputs across different business scenarios to assess whether the generated estimates were reasonable.
- Tested the business-rule alert system under normal, high-risk, and critical business scenarios.
- Verified alert generation and severity classification for different predefined business conditions.

- Tested integrated application workflows involving preprocessing, revenue prediction, and business-rule alerts.
- Investigated authentication-related issues, including HTTP 401 responses encountered during application usage and integration.
- Investigated database schema mismatches and application configuration issues affecting the expected behaviour of certain workflows.
- Investigated local Python, Node.js, frontend environment, dependency, and project version issues encountered during development and testing.
- Reviewed project configuration, dependencies, database structures, and environment settings while troubleshooting identified issues.

3.3.5 Challenges and Resolutions

Data Quality and Preparation: Business datasets required validation for missing values, duplicate records, inconsistent formats, incorrect data types, and unusual values. These issues were addressed through systematic data inspection, preprocessing, exploratory analysis, and validation before the datasets were used in further workflows.

Revenue Prediction and Integration: Revenue prediction required preparing numerical and categorical variables, selecting relevant features, training the model, and integrating the predictive workflow with the application. These requirements were addressed through data preprocessing, encoding, feature preparation, model training, evaluation, and application-level testing.

Business Alert Configuration: The alert system needed to distinguish between normal business activity and situations requiring attention. This challenge was addressed by defining appropriate business thresholds, configuring severity levels, and testing normal, high-risk, and critical business scenarios.

Integration and Version Control: Differences between individual branches and integrated project versions occasionally caused functionality to appear missing or inconsistent. These issues were addressed by reviewing branch versions, checking integration status, and verifying the correct project configuration and implementation.

Authentication and Environment Issues: Authentication errors, database schema mismatches, and local development environment issues were encountered during development and testing. These issues were investigated by reviewing application configuration, database structures, dependencies, environment settings, and project versions.

3.4 Pallavi D R

Pallavi D R contributed to MarketMind AI through frontend and backend setup, frontend-backend integration, user interface improvements, project structure refactoring, Customer Churn Prediction development, testing, and model validation.

3.4.1 Milestone 1 Contributions

During Milestone 1, Pallavi focused on establishing the initial application structure and enabling communication between the major frontend and backend components.

- Contributed to the initial frontend setup and development of the application's user interface structure.
- Contributed to the initial backend setup required to support the application's core functionality.
- Worked on frontend-backend integration to establish communication between the application interface and backend services.
- Assisted in connecting frontend components with backend functionality to support the flow of data, application requests, and responses.
- Contributed to establishing the initial integration workflow between different application components.
- Supported the development of a structured foundation for subsequent features and modules developed during later milestones.

3.4.2 Milestone 2 Contributions

During Milestone 2, Pallavi focused on improving the application's user interface, reorganising the project structure, implementing role-based user functionality, and supporting the continued integration of frontend and backend functionality.

- Enhanced the user interface to improve the overall appearance, usability, and user experience of the application.
- Worked on improving the organisation and structure of frontend components.
- Contributed to refactoring the project structure to make the application more organised, maintainable, and easier to work with.

- Implemented role-based user functionality to support different categories of users within the application.
- Worked on role selection and role-specific user access as part of the application's role-based functionality.
- Assisted in improving the overall user experience through UI-related enhancements and improvements to the application interface.
- Reviewed the interaction between frontend components and backend functionality while working on application improvements.
- Supported the continued integration of implemented features across the frontend and backend.
- Assisted in ensuring that frontend requests and backend responses continued to work together as part of the application workflow.

3.4.3 Milestone 3 Contributions

During Milestone 3, Pallavi focused on developing the Customer Churn Prediction module, analysing customer behaviour, implementing churn risk classification, and resolving a data-related issue that affected prediction accuracy.

- Developed the Customer Churn Prediction module using Logistic Regression.
- Worked on analysing customer behaviour using available customer and transaction-related information.
- Developed functionality for identifying customers based on their predicted churn risk.
- Implemented risk classification to categorise customers according to their likelihood of churn.
- Contributed to generating personalised recommendations based on customer risk and behavioural analysis.
- Tested the churn prediction workflow using different customer scenarios and available application data.
- Identified an issue where multiple sales made by the same customer on the same day were incorrectly treated as separate customer visits.

- Analysed the transaction data to identify how the incorrect visit calculation affected customer behaviour analysis and prediction results.
- Resolved the issue by combining multiple transactions made by the same customer on the same day into a single customer visit.
- Retested the churn prediction workflow after applying the correction to verify that the prediction results were more consistent.

3.4.4 Milestone 4 Contributions

During Milestone 4, Pallavi focused on testing and validating the implemented application features, frontend-backend communication, and Customer Churn Prediction functionality.

- Tested implemented user interface functionality and verified the behaviour of major frontend features.
- Tested frontend-backend integration and communication between application components.
- Reviewed application requests and backend responses to ensure that integrated functionality worked correctly.
- Tested different Customer Churn Prediction scenarios to validate the implemented prediction workflow.
- Verified the churn prediction workflow after resolving the same-day transaction aggregation issue.
- Evaluated the Customer Churn Prediction model using Accuracy, Precision, Recall, and F1 Score.
- Used Stratified Cross-Validation to further evaluate the consistency and reliability of the churn prediction model.
- Reviewed model results and prediction outputs during the testing and validation process.

3.4.5 Challenges and Resolutions

Customer Visit Calculation: An issue was identified where multiple sales made by the same customer on the same day were treated as separate visits. This affected customer

behaviour analysis and resulted in incorrect churn predictions. The issue was resolved by combining same-day transactions for the same customer into a single customer visit before processing the churn prediction workflow.

Frontend-Backend Integration: Establishing and maintaining communication between frontend and backend components required proper coordination between application requests and backend functionality. This was addressed through integration work, continued testing, and verification of the overall application workflow.

Model Validation: The Customer Churn Prediction module required evaluation across different customer scenarios to verify the reliability of its predictions. Model performance was assessed using Accuracy, Precision, Recall, F1 Score, and Stratified Cross-Validation to validate the implemented prediction workflow.

3.5 Pramodh Pillitla

Pramodh Pillitla contributed to MarketMind AI across full-stack development, backend API engineering, database architecture, authentication and role-based access control, AI/ML integration, frontend development, Git-based project integration, debugging, testing, and application stabilisation.

3.5.1 Milestone 1 Contributions

During Milestone 1, Pramodh contributed to establishing the technical foundation of the application, including backend development, authentication, database configuration, frontend structure, and initial application integration.

- Contributed to the development of the FastAPI backend and the initial application architecture.
- Worked on authentication-related backend routes, user registration, backend validation, token generation, and JWT-based authentication for secure user sessions and protected application functionality.
- Contributed to migrating the database configuration from SQLite to PostgreSQL using environment-based configuration, including `DATABASE_URL` and environment variables for database connectivity.
- Contributed to the initial dashboard, frontend application structure, layout development, and navigation.

- Worked on integrating frontend components with backend APIs during the initial stages of application development.
- Developed and supported initial backend functionality for inventory and sales-related workflows.
- Contributed to defining and implementing database models and relationships required for managing users, customers, products, sales, and other core application data.
- Assisted in configuring the backend project structure, API routing, and environment settings to provide a stable foundation for subsequent feature development and integration.

3.5.2 Milestone 2 Contributions

During Milestone 2, Pramodh focused on strengthening the database architecture, multi-tenant structure, role-based access control, backend APIs, business workflows, frontend integration, and application data management.

- Contributed to developing a multi-tenant architecture for separating business-related data and worked with the `business_id` structure across major application entities.
- Contributed to database schema development, Alembic migration workflows, and schema updates involving users, customers, inventory, sales, invoices, categories, suppliers, teams, and datasets.
- Supported database migration and schema synchronisation during application development.
- Worked on implementing Role-Based Access Control (RBAC) and protecting backend endpoints based on user roles and permissions.
- Worked with role-related information within JWT tokens and corrected JWT role claim issues to ensure proper authentication and authorization.
- Contributed to frontend protected routes, role-based access, and authentication state and session management using backend authentication tokens.
- Developed and maintained backend APIs for inventory management, including CRUD operations, stock monitoring, and inventory workflows.
- Contributed to CSV-based data import functionality for inventory and other business-related data.

- Worked on sales and POS transaction processing, including manual sales transactions and batch transaction imports.
- Contributed to automatic inventory stock updates after completed sales transactions and connected sales processing with automated invoice generation.
- Worked on backend functionality for categories, suppliers, users, teams, datasets, and reusable analytics and KPI-related workflows.
- Supported frontend-backend integration and tested authentication, inventory, sales, and related application workflows.

3.5.3 Milestone 3 Contributions

During Milestone 3, Pramodh focused on AI/ML integration, product recommendation workflows, notifications and business alerts, frontend improvements, and integration of major project modules.

- Contributed to developing a reusable machine learning inference structure for loading trained models and generating predictions.
- Worked on integrating machine learning functionality with backend APIs instead of keeping AI models as separate standalone implementations.
- Contributed to backend analytics endpoints for forecasting and churn-related prediction functionality.
- Worked on integrating forecasting, churn prediction, anomaly detection, recommendation, and other AI-related functionality through the central application router.
- Worked on integrating the product recommendation functionality with backend APIs and application workflows.
- Integrated the Online Retail dataset and supporting data preparation workflows required for recommendation functionality.
- Contributed to database seeding and supporting scripts for retail transaction data.
- Worked on connecting recommendation requests with the customer interface and refactored the workflow to support asynchronous API requests and responses.
- Contributed to displaying AI-generated product recommendations within the customer-related application interface.

- Contributed to implementing notification and inventory alert models, structured notification responses, low-stock alerts, and business alert workflows.
- Worked on connecting notification functionality with the frontend, application layout, and notification display interface.
- Contributed to frontend development, including the main application layout, responsive sidebar navigation, application routing, and protected routes.
- Worked on authentication state management and improvements to Login and Registration interfaces.
- Contributed to frontend pages related to datasets, teams, categories, suppliers, inventory, sales, customers, and settings.
- Fixed frontend data mapping issues affecting sales and inventory information and contributed to customer interface improvements and AI recommendation integration.
- Performed frontend code cleanup and resolved lint warnings, React warnings, undefined variables, and unused code.
- Tested AI recommendation integration, asynchronous API requests, authentication workflows, inventory and sales functionality, and frontend-backend communication.

3.5.4 Milestone 4 Contributions

During Milestone 4, Pramodh focused on integrating parallel development work, managing the shared Git workflow, debugging integration issues, stabilising the application after merges, configuring the development environment, testing major application workflows, and resolving issues across the frontend, backend, database, authentication, and AI/ML components.

- Played a significant role in integrating parallel development work into the shared project branch using Git and GitHub.
- Worked with the shared **pre-dev** branch and coordinated the integration of contributions from multiple team member branches into the combined application.
- Reviewed changes from different branches before integration to identify potential conflicts and dependencies between independently developed features.
- Resolved Git merge conflicts involving frontend components, backend services, authentication logic, database-related files, and AI/ML functionality.

- Ensured that integrated changes remained compatible with the existing project structure and did not unnecessarily overwrite previously implemented functionality.
- Performed post-merge debugging to identify issues that appeared only after combining contributions from multiple branches.
- Investigated missing functionality, schema differences, version mismatches, and other integration-related problems introduced during collaborative development.
- Contributed to resolving database schema mismatches and missing database column issues encountered after branch integration.
- Investigated backend router conflicts and application startup failures that occurred during the integration and testing process.
- Debugged backend functionality to identify errors in API execution, database interactions, and communication between different application modules.
- Worked on frontend stability by resolving React `useEffect` warnings, undefined variables, dead code, and lint-related issues.
- Investigated frontend-backend request and response mismatches and resolved HTTP 422 errors caused by differences between registration payloads and backend validation requirements.
- Fixed JWT role-related issues affecting authentication and authorization workflows and helped ensure that authenticated users were correctly handled according to their assigned roles.
- Tested authentication and authorization flows after integration to verify that protected application functionality continued to work correctly.
- Tested inventory and sales workflows after frontend-backend integration and corrected data mapping issues affecting the display of sales and inventory information.
- Verified that frontend components correctly consumed backend API responses after changes were merged from different development branches.
- Performed testing of AI recommendation integration and verified that recommendation-related functionality continued to operate correctly after branch merges and subsequent bug fixes.
- Investigated integration issues between AI/ML components and the application backend and contributed to resolving issues affecting their execution.

- Tested major application workflows after integration to identify regressions and functionality that had been affected by changes from other branches.
- Repeatedly tested and debugged the application after fixes to confirm that previously identified integration issues were resolved without introducing additional problems.
- Contributed to environment configuration by preparing an `.env.example` file containing placeholders for database, JWT, and SMTP-related configuration settings.
- Supported PostgreSQL environment configuration and project setup required for running the integrated application.
- Worked on configuration-related issues encountered while connecting application components to the configured database environment.
- Helped stabilise the integrated application by identifying and resolving issues across multiple layers instead of treating frontend, backend, database, and AI/ML problems independently.
- Maintained the shared project branch during the integration process and ensured that fixes and validated changes were incorporated into the collaborative development workflow.
- Contributed to final application improvements by testing integrated features, correcting implementation issues, and improving the overall stability of the application.
- Supported the team in maintaining a functional and consistent version of the project after multiple parallel contributions had been integrated.

3.5.5 Challenges and Resolutions

Authentication and Registration Issues: Frontend and backend authentication workflows occasionally contained differences in expected request data. A registration-related HTTP 422 error was investigated and resolved by synchronising the frontend request payload with backend validation requirements and the required tenant information.

JWT Role and Authorization Issues: A JWT role claim issue affected the availability of correct user role information during authentication. This was resolved by correcting the role information included within authentication tokens and verifying role-based access behaviour.

Database Schema Synchronisation: Integration between different development branches occasionally resulted in missing database columns and schema mismatches. These issues were addressed by reviewing database migrations, synchronising schemas, and correcting affected application code.

Frontend-Backend Data Mapping: Some sales and inventory information was not displayed correctly because of differences between backend response structures and frontend data mappings. These issues were resolved by correcting the relevant mappings and testing the affected application pages.

Branch Integration Conflicts: Parallel development across multiple team branches resulted in conflicts between frontend, backend, database, and AI/ML components. These issues were addressed by reviewing branch differences, resolving conflicts, merging the required changes, and performing post-merge testing.

Frontend Stability and Code Quality: React warnings, undefined variables, unused code, and lint issues affected application stability during development. These issues were resolved through code cleanup, debugging, and repeated testing of the affected components.

Post-Merge Application Stability: Differences between individual branches occasionally caused missing features, router conflicts, application crashes, and inconsistent functionality after integration. These issues were investigated through post-merge debugging, version comparison, backend and frontend verification, and repeated integration testing.

3.6 Sathish

Sathish contributed to the database architecture, product recommendation system, backend API integration, and performance optimisation of MarketMind AI.

3.6.1 Milestone 2 Contributions

- Contributed to the design and refinement of relational database models using SQLAlchemy ORM.
- Worked on database connectivity and session management for PostgreSQL and SQLite environments.
- Supported the implementation of primary keys, foreign keys, indexes, and relationships between application entities.

- Contributed to Alembic-based database migrations for managing schema changes and AI-related database tables.
- Supported multi-tenant data isolation by maintaining appropriate business-level database relationships and query scoping.

3.6.2 Milestone 3 Contributions

- Contributed to the development of the product recommendation system using customer purchase and product transaction data.
- Implemented co-purchase and association-based recommendation logic for identifying complementary products.
- Supported category-affinity and popularity-based recommendation strategies for cases with limited transaction history.
- Contributed to cross-selling, up-selling, and cold-start recommendation functionality.
- Integrated recommendation services with FastAPI endpoints and supported their consumption through the React frontend.
- Contributed to ranking and filtering recommended products based on customer purchasing behaviour and product relevance.
- Tested recommendation outputs across different customer scenarios and supported debugging and refinement of the recommendation workflow.

3.6.3 Milestone 4 Contributions

- Contributed to testing and validation of major application features and backend API functionality.
- Supported error handling and debugging of issues identified during application testing.
- Worked on identifying and resolving integration-related issues between frontend components and backend APIs.
- Supported validation of API responses, application workflows, and overall system functionality.
- Contributed to final system integration, testing, and documentation activities.

- Assisted in validating database interactions and ensuring that application data was correctly retrieved, processed, and displayed across integrated workflows.
- Supported testing of authentication and authorization workflows to verify that users could access application features according to their assigned roles and permissions.
- Performed regression testing after fixes and integration changes to verify that existing application functionality continued to work correctly.

3.6.4 Challenges and Resolutions

- **Challenge:** Errors and inconsistencies were identified during frontend, backend, and API integration testing.
- **Resolution:** Issues were debugged and resolved through repeated testing, validation, and correction of integration and API-related problems.
- **Challenge:** Ensuring that the integrated application remained stable across different workflows.
- **Resolution:** Major workflows and application features were tested and validated before final integration.

Chapter 4

Major Challenges and Resolutions

During the development and integration of MarketMind AI, the team encountered several technical, data-related, integration, and collaboration challenges. These challenges were addressed through systematic debugging, testing, validation, code integration, and coordination among the team members.

4.1 Frontend and Backend Integration

One of the major challenges during development was maintaining consistent communication between the React frontend and FastAPI backend. As multiple features were developed and integrated independently, differences in API response structures, request payloads, field names, and application workflows occasionally resulted in incorrect data display or frontend errors.

These issues were addressed by reviewing the frontend requests and backend responses, identifying mismatches between the expected and actual data structures, and updating the affected components accordingly. The team also performed repeated integration testing to verify that data was correctly transferred between the frontend, backend services, database, and AI/ML modules.

Additional frontend stability issues, including React warnings, undefined variables, loading problems, and crashes caused by changes in backend response formats, were resolved through code cleanup, improved data handling, loading states, error handling, and repeated testing.

4.2 Data Processing and AI/ML Challenges

The project involved multiple datasets related to customers, sales, products, inventory, and business transactions. Preparing these datasets for analytics and machine learning required handling missing values, duplicate records, inconsistent formats, incorrect data types, and unusual values.

Data preprocessing and validation were therefore performed before the datasets were used by the AI/ML workflows. Exploratory analysis and feature preparation were also carried out to ensure that relevant information was available for customer segmentation, sales forecasting, churn prediction, and product recommendation.

AI/ML workflows also required validation across different input scenarios. Customer segmentation, sales forecasting, churn prediction, and product recommendation outputs were reviewed to identify unexpected behaviour and ensure that the integrated models and workflows produced consistent results.

A specific data-related issue was identified in the Customer Churn Prediction workflow, where multiple transactions made by the same customer on the same day were incorrectly treated as separate customer visits. This affected customer behaviour analysis and churn predictions. The issue was resolved by aggregating same-day transactions for the same customer into a single visit before processing the prediction workflow.

Model evaluation was also performed using appropriate performance measures. Depending on the AI/ML workflow, evaluation involved metrics such as Accuracy, Precision, Recall, F1 Score, MAE, RMSE, and R^2 score, along with validation approaches such as Stratified Cross-Validation.

4.3 API and Communication Issues

API communication and application connectivity presented additional challenges during development and testing. Issues included incorrect request payloads, backend validation errors, authentication failures, and CORS-related communication problems between the frontend and backend.

A registration-related HTTP 422 error occurred because the data submitted by the frontend did not match the validation requirements expected by the backend. The issue was resolved by synchronising the frontend request payload with the backend schema and required application information.

Authentication and authorization issues were also encountered, including JWT role-related problems and HTTP 401 responses. These were investigated by reviewing au-

thentication tokens, role information, protected routes, and backend authorization requirements. The authentication workflow was updated and tested to ensure that users received the appropriate access based on their roles and permissions.

CORS issues affected communication between frontend development origins and the backend during local testing. The application configuration was reviewed and updated to allow the required development origins. The affected APIs and frontend workflows were then tested again to verify successful communication.

Database connectivity and schema-related issues were also investigated when API workflows depended on database structures that differed between integrated development versions. The team reviewed database configurations, migrations, schemas, and application code to identify and address these inconsistencies.

4.4 Git and Version Control Challenges

Since MarketMind AI was developed collaboratively, multiple team members worked on separate branches and contributed changes to interconnected frontend, backend, database, and AI/ML components. This resulted in branch differences, merge conflicts, and situations where functionality appeared differently across development and integrated versions.

The team addressed these challenges by reviewing branch histories and differences, synchronising branches, manually resolving conflicts, and integrating only the required changes. Post-merge testing was performed to identify missing functionality, router conflicts, schema differences, and other integration problems.

Particular attention was required when multiple developers modified common application files. Changes were reviewed before integration to reduce the possibility of overwriting existing functionality. The team also used GitHub branches and pull/merge workflows to maintain a controlled collaborative development process.

4.5 Testing and Debugging Challenges

Testing the complete MarketMind AI application required validating individual modules as well as their interactions with the rest of the system. The team tested frontend components, backend APIs, database workflows, authentication, CRUD operations, AI/ML modules, and integrated application functionality.

During testing, several issues were identified, including frontend crashes, missing routes, undefined variables, React warnings, incorrect data mappings, authentication errors, database schema mismatches, and inconsistent behaviour after branch integration.

These issues were addressed through iterative debugging and retesting. Meaningful logging and improved error handling were used to make application issues easier to identify. Loading states, empty states, fallback behaviour, and error boundaries were also introduced or improved where required.

The integrated AI/ML workflows were tested using different scenarios and inputs to verify their behaviour after integration. Frontend and backend requests were monitored to ensure that APIs returned the expected information and that the corresponding frontend components displayed the results correctly.

Overall, repeated testing, debugging, validation, and cross-module integration helped improve the stability and reliability of the final MarketMind AI application.

Chapter 5

Conclusion

This document summarizes the individual and collaborative contributions made by the team members during the development of the MarketMind AI project.

The project involved contributions across multiple areas, including frontend development, backend development, database management, AI/ML implementation, data processing, API integration, testing, debugging, version control, and documentation.

Through collaborative development and integration of individual modules, the team worked toward building a unified AI-powered business intelligence and decision support platform.